NYP CNC AI Chatbot Helper

By Ilysia (232734X) and Bryan

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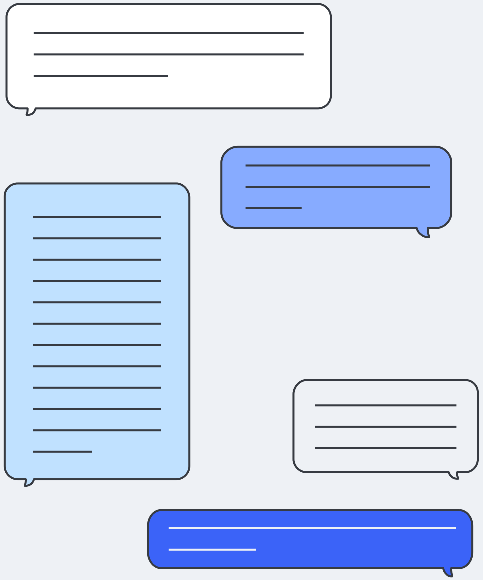


01

PROJECT BRIEF

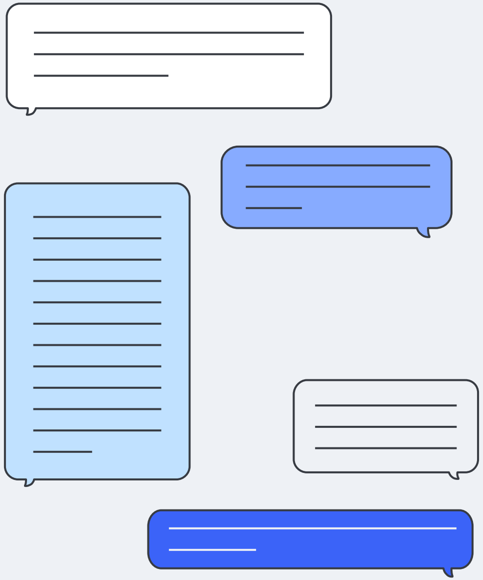
Project Overview

Aim

To build an AI chatbot that is context specific to the NYP Data Classification Systems.

Purpose

To facilitate and answer potential queries staff have about different CNC related questions, instead of having to search and refer to the documentation files to find the answers.



02

BRYAN'S PART

Timeline

Wk 3 — **Sprint 1**

- Project Understanding
- Digesting the high level flow of the project
- Experimenting to find out portions that can be improved

Wk 6 — **Sprint 2**

- Implementing the unstructured library on an improvement on text analysis capabilities
- Conducting tests on the library's accuracy

Wk 9 — **Sprint 3**

- Implementing the metadata tagger
- Testing new chunking methods such as semantic chunking for smarter chunking
- Implementing the new technologies

Wk 12 — **Sprint 4**

- Figuring out how to package the dependencies
- Final Report
- Final Presentation Slides

Data Processing Logic Old

First Batch

Data Initialization

1. Import multiple libraries to extract text from docx, powerpoint, pdf and excel
2. Extract all the text and compile them into one text file
3. Ingest the text file
4. Transform it from string to langchain's object Documents
5. Chunking using a user set recursive text splitter
6. Batching for better processing
7. Creating a database from the chunks

Data Upload

Same logic as Data Initialization

Second Batch

Data Initialization

1. Import multiple libraries to extract text from docx, powerpoint, pdf and excel
2. Extract all the text and compile them into one text file
3. Clean all the hidden unicode characters as well as anything that is not alphabetic or spaces
4. Chunk using splicing
5. Remove duplicates in chunking
6. Insert chunks into the database

Data Upload

1. Using Data Initialization code to extract the text
2. Chunk using splicing
3. Upload chunks into the database
4. File is sorted and stored by file type

Data Processing Logic New

Data Initialization

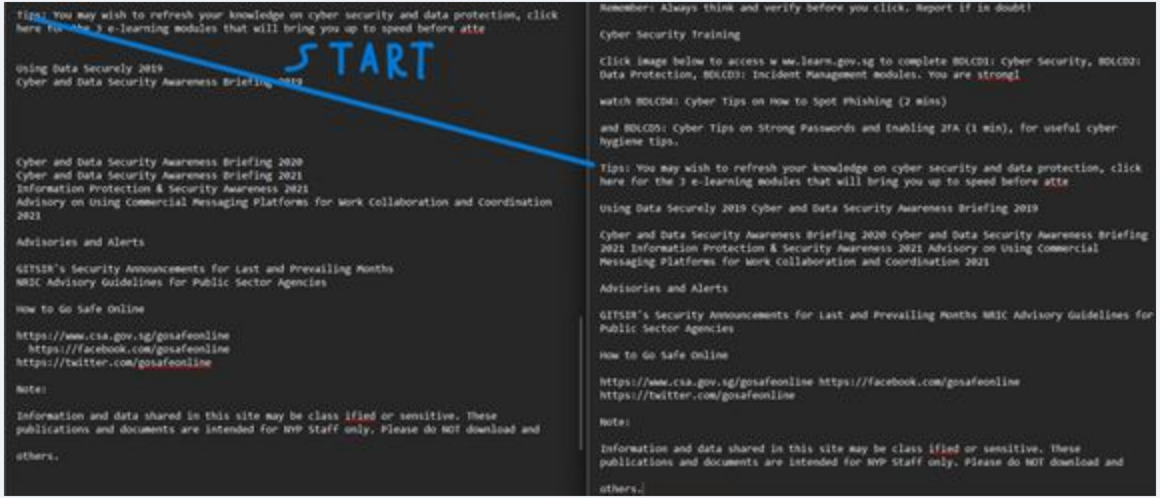
1. Import the unstructured library which automatically processes files
2. Initialize two different collections in chroma, one for classification and one for chatbot
3. Each file is processed individually
4. Text is extracted from the file
5. It is tagged with a metadata tagger
6. It is chunked using a semantic chunker
7. Batched for efficiency
8. Text is pushed into the appropriate vector database

Data Upload

1. Data is processed same as initialization
2. File is sorted and stored by file type

Summary of Unstructured Library Demo PDF

PDFs, Generally both will output a similar result as both unstructured and the previous batch uses pdfminer, but there are slight changes in the spacing as well as in the order of the scraped text



Documents

Documents, both are able to extract the text but not the image attached to the document.

Dear Students,

Do you want to showcase your ideas while positively impacting society?

Form a team of 3 to 5 and challenge yourselves to develop technological solutions to address prevalent social issues in Singapore.

Organised by Toa Payoh East Youth Network, Marymount Youth Network and Ngee Ann Polytechnic's Engineering Interest Group (EIG), ideateCOMM Challenge provides a platform for youths to demonstrate their innovative solutions to prevalent social issues.

Why should you join? Here's 3 simple reasons.
Plethora of industry relevant workshops:
Sales & Marketing | Digital Fabrication | UI/UX with Figma and Wix
Attractive Prizes & Networking Opportunities:
Stand a chance to win prizes of up to $2,000 shopping vouchers and network with industry experts.
Positively impact society:
Our partners intend to execute feasible and innovative ideas so that you will have the chance to impact the world for real.

Registration is FREE! Join ideateCOMM Challenge NOW!

REGISTER & FIND OUT MORE:
Visit the official website https://www.ideatecomm.com for more details on the competition, and follow us on Instagram: @ideate.comm for more updates. Please contact ideateCOMM organising committee at ideatecomm@gmail.com if you have any enquiries about the competition.

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Excel

While the extracted text is the same, unstructured extracts tables as one chunk, this is not limited to excel

```
Data Stock-taking Summary Sheet

Please indicate the datasets that are regularly used by your department / school.
Examples of usage: 1. for annual submission to MOE, SSG, PSD etc
2. for management reporting
3. for publicity or for public information

Owner Dataset ID Dataset Name Usage Shared to Frequency
HR Age-Group and Length-Of-Service Profile
QPO NYP-KPI-LST-001 NYP_INDICATORS_KPI_DEFINITIONS NYP Management Dashboard NYP annual
HR NYP-KPI-EES-001 NYP_INDICATORS_KPI_EES NYP Management Dashboard NYP annual
RGO LEARNER-ADM-INT-002 Nanyang Polytechnic Student Enrolment, Annual Data.gov.sg Public
annual
RGO LEARNER-ADM-PIT-001 Nanyang Polytechnic Planned Intake, Annual Data.gov.sg Public
annual
RGO LEARNER-ADM-INT-001 Nanyang Polytechnic Student Intake, Annual Data.gov.sg Public
annual
RGO Course-PET-DF-001 Full-time Diploma Courses Data.gov.sg Public annual
CIL3 Course-LLL--All-001 Lifelong Learning Courses for Adult Learners Data.gov.sg Public
annual
RGO LEARNER-ADM-LAS-001 Nanyang Polytechnic GCE 'O' Level Aggregate Cut-Off_Points by
Course Data.gov.sg Public annual
RGO LEARNER-GES-FG-2020 NYP GES 2020 for Graduates Class 2020 MOE Policy Analysis
NYP Management Review GES Contractor
MOE annual
RGO LEARNER-GES-PNS-2020 NYP GES 2020 for Post NS Class 2017 MOE Policy Analysis
NYP Management Review GES Contractor
MOE annual
RGO LEARNER GES-PG-2020 NYP GES 2020 for Poly Graduates Class 2015 MOE Policy Analysis
NYP Management Review GES Contractor
MOE annual
RGO LEARNER-WSPOS-PPT-2020 NYP Work Study Programme Outcome Survey 2020 - Participants SSG
Policy Analysis
NYP Management Review WSPOS Contractor
SSG annual
RGO LEARNER-WSPOS-EMP-2020 NYP Work Study Programme Outcome Survey 2020 - Employers SSG
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RGO LEARNER-WSPOS-PPT-2020 NYP Work Study Programme Outcome Survey 2020 - Participants SSG
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EMP-2020 NYP Work Study Programme Outcome Survey 2020 - Employers SSG Policy Analysis\nNYP
Management Review WSPOS Contractor\nSSG annual HR STAFF-PLAN-PSD-001 Staff Strength and
Movement Annual returns for analysis PSD annual HR STAFF-PLAN-MOM-001 Labour Market Survey
MOM Policy Analysis MOM annual


Owner Dataset ID Dataset Name Date Created Last Updated Description Classification
Frequency Coverage Start Coverage End Used By Subject Area HR Age-Group and Length-Of-
Service Profile To show the distribution of various age groups and length of service
Classified – OC_NS Annual 1992-04-01 00:00:00 2019-10-18 00:00:00 Staff QPO NYP-KPI-LST-001
NYP_INDICATORS_KPI_DEFINITIONS 2020-11-30 00:00:00 2020-11-30 00:00:00 KPI Definitions.
Used in NYP Management Dashboard (NYPMD) Classified – OC_NS Ad-hoc 2019-01-01 00:00:00
```

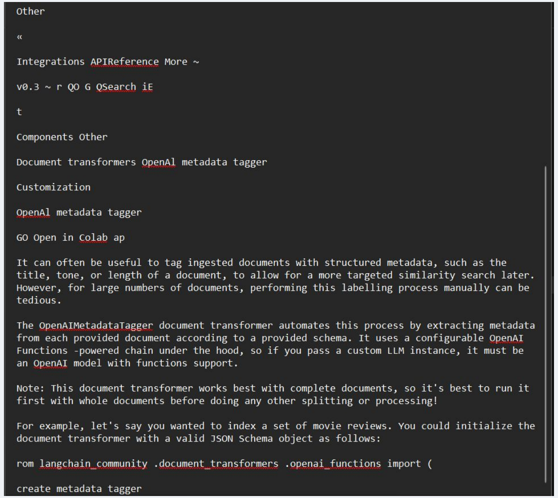
Powerpoint

For Powerpoint, there are further spacing differences but the content of the scraped text is the same, the one on the left is batch 1 and the one on the right is unstructured.

Images

Images was where it had the most difficulties,

- it interpreted symbols such as company logos as nonsensical letters and symbols,
- it interpreted some capital letter I as the | symbol
- as well as missing out on other stuff,
- whereas for screenshots of code blocks with numbered lines it does not preserve the code and the numbers on the same line
- but it can interpret a line that was cut in half



Other

«

Integrations APIReference More ~

v0.3 ~ r QO G QSearch iE

t

Components Other

Document transformers OpenAl metadata tagger

Customization

OpenAl metadata tagger

GO Open in Colab ap

It can often be useful to tag ingested documents with structured metadata, such as the title, tone, or length of a document, to allow for a more targeted similarity search later. However, for large numbers of documents, performing this labelling process manually can be tedious.

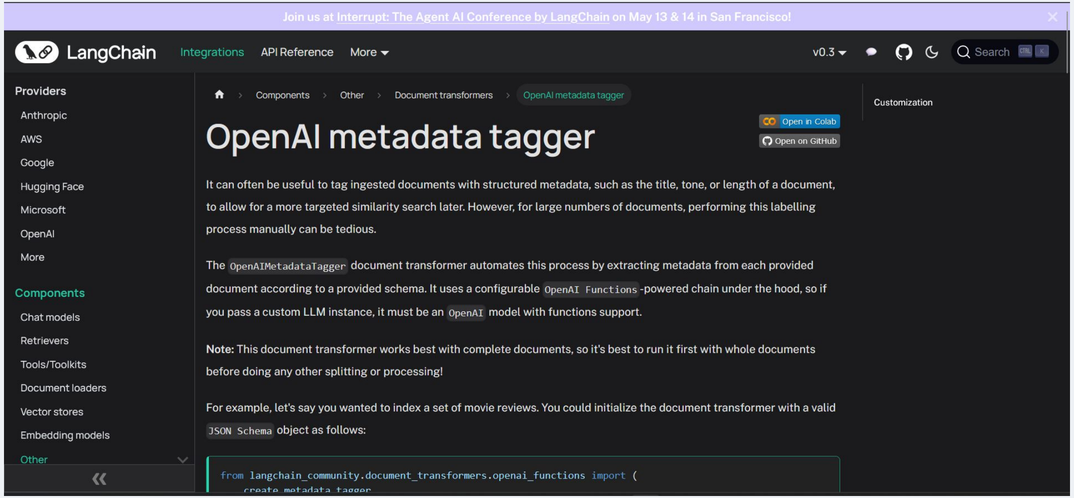
The OpenAIMetadataTagger document transformer automates this process by extracting metadata from each provided document according to a provided schema. It uses a configurable OpenAI Functions -powered chain under the hood, so if you pass a custom LLM instance, it must be an OpenAI model with functions support.

Note: This document transformer works best with complete documents, so it's best to run it first with whole documents before doing any other splitting or processing!

For example, let's say you wanted to index a set of movie reviews. You could initialize the document transformer with a valid JSON Schema object as follows:

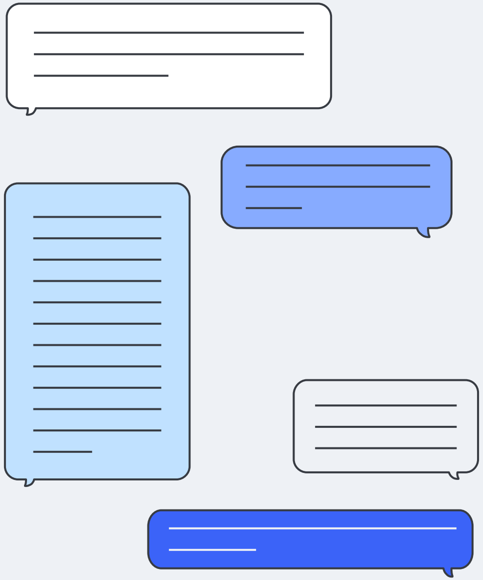
rom langchain_community .document_transformers .openai_functions import (

create metadata tagger



Potential Improvements

1. By separating classification and chatbot data, we can increase the classification speed higher by cutting down on the irrelevant datasets that the algorithm has to sift through
2. While the second batch's chunking method was far too rigorous causing certain information to get lost. I.e. There were a dataset which a flowchart of questions if this go to question 2, if that go to question 3. Therefore we will revert back to using langchain's text splitter but switching over to semantic chunking.
    a. Cuts down on the amount of chunks (recursive text splitter with overlap meant that each chunk has only 200 unique characters, which the other 800 characters are overlap with the other chunks)
    b. Semantic chunking makes it so each chunk is mostly self contained
3. The added metadata tagger makes retrieval faster through the use of keyword tags to identify important chunks



03

DEPLOYMENT

01 Review

Review Past Chatbots + Approach

02 Evaluate

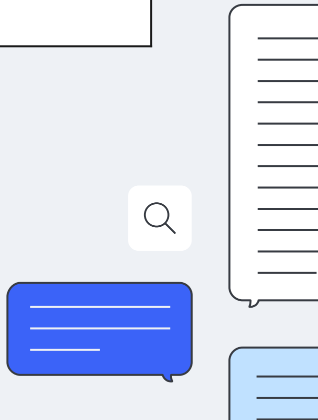
Comparison of models

03 Development

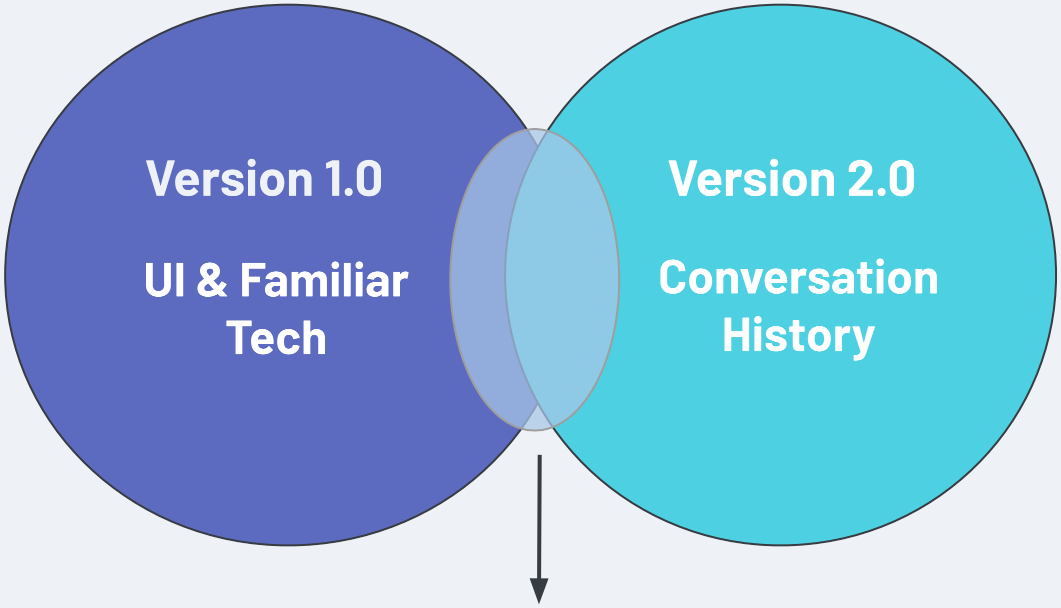
Introduce Version 3.0

Review of Previous Chatbots

	Chatbot V1.0	Chatbot V2.0
UI	✅ Clean & Intuitive	❌ Unclear buttons, cluttered
Functions	❌ No conversation history	✅Conversation history added
Language	✅Python, Flask, Streamlit	❌Docker, Svelte (unfamiliar)



Project Approach



Version 3.0

Best of both worlds

Timeline

Wk 3 — Wk 6 — Wk 9 — Wk 12

Sprint 1

- Project Understanding
- Evaluate Previous UI
- Completed 1st Prototype

Sprint 2

- Compare vector databases
- Speech-to-text Function
- Testing the accuracy of the Speech-to-text function

Sprint 3

- Improve the UI
- Redoing the conversation history using Flask
- Testing the accuracy
- Integration with Bryan's Part
- Final Report

Sprint 4

- Final Report
- Final Presentation Slides

Vector Database Comparison

	ChromaDB (Used in V1.0)	PostgreSQL (Used in V2.0)
Speed	✅ Fast vector search	❌ Slower for large queries
Scalability	✅ Optimized for AI workloads	⚠️ Scalable with tuning
Accuracy	✅ Efficient ANN search	❌ Requires vector extensions for optimal search
Persistence	❌ Limited structured storage	✅ Robust, data storage & history
Integration	✅ Easy to set up for AI apps	❌ Requires Pgvector extension

Chosen: ChromaDB

Speech-to-Text (STT)

STT Model Comparison

	GPT-4o	GPT-4o Mini
NLP Performance	Advanced reasoning & long context	Solid for general queries
Latency	❌Slower (high compute)	✅Fast & Responsive
Cost	❌High	✅Low
Voice Handling	✅Native multimodal support (voice)	❌ Text input/output
Best For	Complex AI agents	Lightweight, cost-effective chatbots

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Chosen: GPT-4o Mini

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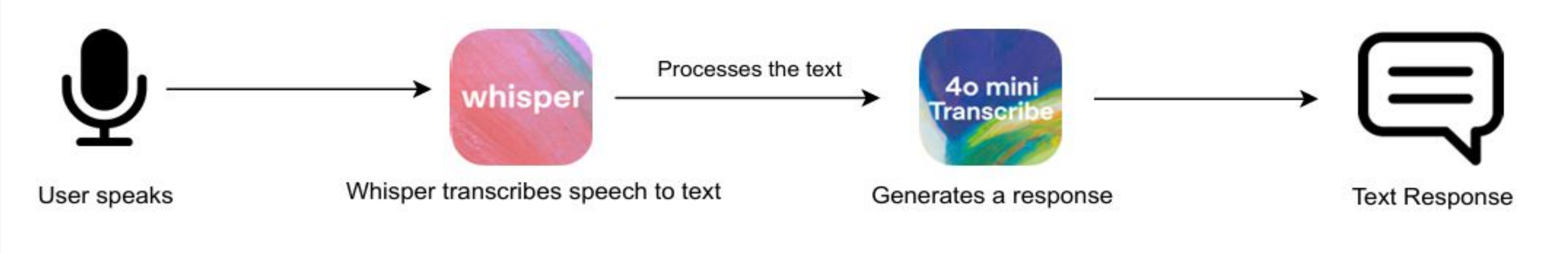
Oh No!

Chosen: GPT-4o Mini

Use Whisper!

What can Whisper do?

- Transcribes voice to text
- Supports multiple languages, noise robustness
- Lightweight and open-source



GPT-4o Mini + Whisper - Cost-effective, Efficient Speech-to-text function

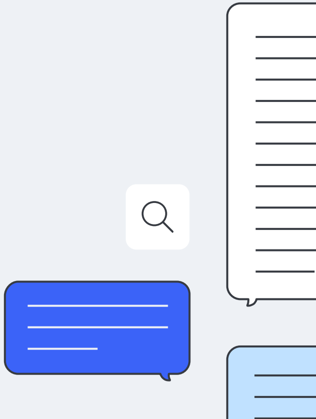
Summary

What I've done so far:

- Speech-to-text function
- Testing for accuracy

Next Steps:

- Implement the model with conversation history
- Testing for accuracy
- Improvements to UI



Chatbot Demo

THANK YOU